

Grade-Aware Smart Word Substitution for Second Language English Writers

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Abstract—Existing AI-powered writing assistants for English as a Second Language (ESL) learners predominantly focus on grammatical error correction or automated text rewriting, offering limited support for deliberate vocabulary development. This project presents a multi-stage NLP pipeline for grade-aware smart word substitution that helps English learners improve vocabulary sophistication in their writing. The system replaces simple or imprecise words with contextually appropriate, grade-level-suitable alternatives using a combination of WordNet synonym generation, Vocabulary Expected Grade Level (VXGL) filtering, RoBERTa-based contextual reranking, BARTScore fluency evaluation, and Sentence-BERT (SBERT) semantic preservation verification. Unlike conventional writing tools that generate or paraphrase entire texts, the proposed pipeline operates strictly at the lexical level: it identifies improvable target words, generates synonym candidates from WordNet, filters them by pedagogical grade-level appropriateness, reranks them for contextual naturalness, and provides brief explanations for each suggestion. By integrating structured linguistic resources with neural scoring models under a non-generative constraint, this system aims to bridge the gap between automated writing correction and effective second language vocabulary acquisition while maintaining learner agency and transparency.

Index Terms—Smart Word Substitution, Vocabulary Grade-Level Filtering, Masked Language Model Reranking, BARTScore, Writing Assistance for ESL Learners

I. INTRODUCTION

The recent breakthroughs in natural language processing have resulted in the widespread adoption of AI-based writing assistants across academic, professional, and personal settings. Grammar checkers, paraphrasing systems, and other automated tools can quickly improve fluency and surface-level correctness, but they typically function as black-box systems that do not encourage a deeper understanding of language. For learners studying English as a second language (ESL), this becomes a serious constraint: errors are rectified, but vocabulary development and in-depth comprehension of lexical choice are often missed.

Empirical research in second language acquisition consistently highlights vocabulary knowledge as a key determinant of writing competence, reading comprehension, and long-term language proficiency. Yet the majority of commercial writing aids concentrate on restructuring sentences, adjusting tone, or fully generating text rather than providing focused vocabulary training. Furthermore, text generation systems that produce content directly pose a threat to learner agency and are likely to foster passive dependency on technology instead of active capacity building.

This project addresses these gaps by developing a grade-aware smart word substitution pipeline that operates strictly at the lexical level. Rather than generating or paraphrasing text, the system identifies generic or imprecise word choices as the user writes, generates synonym candidates from WordNet, filters them by vocabulary expected grade level (VXGL) to ensure pedagogical appropriateness, reranks candidates using RoBERTa contextual probabilities and BARTScore fluency evaluation, and verifies semantic preservation through Sentence-BERT similarity. This multi-stage design ensures that each suggestion is simultaneously contextually natural, semantically faithful, and grade-appropriate for the target learner population.

The proposed system, unlike existing tools, does not produce original content, paraphrase whole sentences, or alter the writer’s intended meaning. Instead, it functions as an intelligent vocabulary tutor embedded within the writing process. By combining structured linguistic resources (WordNet, VXGL) with neural scoring models (RoBERTa, BARTScore, SBERT) under a strict non-generative policy, the pipeline supports vocabulary learning through real-time, explanation-driven feedback without sacrificing user control over the writing process.

II. PROBLEM STATEMENT

The principal problem this project addresses is that existing AI writing aids for second language English learners do not provide grade-aware, pedagogically grounded feedback on vocabulary choice. Current tools either focus on grammatical precision or offer general vocabulary suggestions without considering the learner’s proficiency level or providing the linguistic reasoning behind the recommended changes. Consequently, users may achieve short-term improvements in writing quality but fail to develop the independent lexical sophistication needed for long-term growth.

Specifically, masked language model (MLM)-based generation approaches tend to predict contextually probable but non-synonymous words, while unfiltered large language model (LLM) prompting lacks explicit grade-level control. Neither approach ensures that substitution suggestions are simultaneously contextually natural, semantically faithful, and appropriate for the learner’s current proficiency level.

We hypothesize that a WordNet-first, multi-model pipeline that generates synonym candidates, filters them by vocabulary expected grade level (VXGL), reranks them using contextual masked language model probabilities (RoBERTa), and validates them through sentence-level fluency scoring (BARTScore) and semantic similarity verification (SBERT) can produce word substitutions that outperform both MLM-only generation approaches and LLM prompting baselines lacking explicit grade-level awareness.

III. MOTIVATION

Vocabulary knowledge is the fundamental aspect of effective written communication, especially for non-native speakers of English in academic and professional settings. The challenge for ESL learners is not a lack of ideas, but an inability to find the right words with the right contextual precision to convey those ideas. Solutions that directly address this lexical gap carry practical value: they can lead to better academic outcomes, greater confidence in writing, and reduced reliance on automated rewriting programs that obscure the learning process.

From a technical standpoint, this project is motivated by a specific limitation observed in current smart word substitution (SWS) approaches. Existing baselines on the SWS benchmark achieve at most an $F_{0.5}$ score of 0.201 (BERT-based), far below human performance of 0.359 [8]. A key reason is the reliance on masked language models for candidate generation, which predict contextually probable words rather than true synonyms. Additionally, no existing SWS system incorporates grade-level awareness, meaning suggestions may be either too simple to promote vocabulary growth or too advanced for the learner’s current proficiency.

In a broader scope, this project contributes to the development of human-centered AI systems that prioritize learning and transparency over automation. By creating a pipeline that explains its actions, enforces grade-appropriate suggestions, and upholds learner agency through a strict non-generative constraint, the project aligns with ethical AI principles and educational best practices. Its practical relevance extends to language education platforms, workplace productivity tools, and accessibility-oriented writing technologies.

IV. BACKGROUND AND RELATED WORK

Milton (2013) examines the extent to which various aspects of vocabulary knowledge—vocabulary size and vocabulary depth in particular—are predictors of overall language proficiency in reading, writing, listening, and speaking, with vocabulary being a significant predictor of performance in all four skills. Milton [1] provides empirical evidence supporting the notion that lexical development is not a peripheral issue but rather a central pathway to enhanced communicative competence. The significance of this work is that vocabulary measures can explain a large proportion of the variance

in proficiency scores, often more than other linguistic sub-components, which means that improving learners’ lexical breadth and depth should directly improve their writing and other skills. A major drawback is that vocabulary size and depth were determined by standardized tests, which may not adequately capture learners’ productive and contextual lexical knowledge in actual communicative contexts. Additionally, the results are derived from a specific group of learners and a specific set of tasks, limiting generalizability across proficiency levels and assessment types.

Ranalli, Link, and Chukharev-Hudilainen (2017) discuss the potential of using an automated writing evaluation (AWE) system for formative assessment in second language writing, investigating whether AWE feedback is accurate and useful to learners and teachers. The article [2] represents one of the most important lines of work concerning AI-based writing support tools that offer corrective feedback on grammar, mechanics, and other surface characteristics, demonstrating both their advantages (improved accuracy, more revisions, increased fluency) and their disadvantages. The article provides motivation to develop more explanation-rich, vocabulary-oriented, real-time assistance that specifically addresses the gaps AWE tools do not fully cover. However, the study was confined to one AWE system (Criterion) and a particular learner group, limiting generalizability across other tools, genres, and instructional settings. Moreover, since accuracy differed significantly among error types and the system focused primarily on surface-level features, the study does not comprehensively represent how AWE may support higher-level features of L2 writing such as global structure and content elaboration.

Koltovskaia (2020) examines the behavioral, cognitive, and affective reactions of ESL university students when using Grammarly’s automated corrective writing feedback to revise their writing. The research indicates that automated feedback does not automatically motivate students to accept or reject suggestions; rather, engagement is determined by trust in the tool, comprehension of the feedback, motivation, and prior experiences with English and technology. This article [3] highlights a fundamental weakness of current automated feedback technologies: even when feedback is available, learners may not intelligibly process or learn from it, particularly when it is opaque or overwhelming. Its relevance to the proposed system lies in the importance of creating feedback that is clear, explanation-rich, and pedagogically friendly, so that learners can actively respond to suggestions and transform them into real vocabulary acquisition. The study is limited, however, by its small multiple case-study sample of two ESL students using a single tool (Grammarly), restricting generalizability to larger L2 learner populations and other feedback systems.

Huang et al. (2023) explore the opportunities of ChatGPT to facilitate second language learning, discussing its possible

roles as a conversational partner, feedback provider, and learning resource generator. This paper [4] contextualizes ChatGPT within L2 education, reporting advantages (rich input, personalized interaction, high availability) as well as disadvantages (accuracy issues, over-reliance, and misalignment with pedagogical goals). Its relevance to the proposed system is in the contrast it provides: unlike ChatGPT, which is a general-purpose generator that can assume much of the writing process, the pipeline in this project is specifically limited to non-generative, explanation-based vocabulary assistance. The research is constrained by its reliance on self-reported perceptions from a particular group of learners in a specific institutional environment, and its exploratory, short-term classroom focus limits evidence concerning long-term language development outcomes.

Warschauer and Tong (2023) address the ways generative AI systems like ChatGPT are transforming writing instruction, noting that these systems create new opportunities to support students while raising important questions about authorship, assessment, and equity. The paper [5] provides a conceptual framework for understanding AI-generated text as supplementary rather than substitutive of core writing processes, and underlines the importance of explicit AI literacy. Its value to the proposed system is that it positions the project within the paradigm of AI-as-support rather than AI-as-replacement: the pipeline does not produce full texts, but provides limited, interpretable lexical assistance that supports rather than replaces the writing and learning of students. The article is primarily conceptual and illustrative, however, lacking empirical data on the actual impact of generative AI on writing development across diverse settings.

Shintani (2016) [6] explores the effectiveness of computer-assisted pronunciation feedback in helping second language learners achieve higher pronunciation accuracy. The paper demonstrates that fine-grained feedback targeting a specific linguistic sub-skill with the help of computers can be pedagogically effective, supporting the broader notion that technology can facilitate learning of particular elements of competence. This is relevant to the proposed vocabulary coaching system in that it provides a parallel: just as pronunciation-focused feedback can enhance accuracy in spoken production, a well-designed vocabulary-focused pipeline can use real-time explicit feedback on lexical choice to develop more profound lexical knowledge in L2 writing. The study is limited by its small learner group and specific instructional situation, and its focus on short-term benefits provides less data on the long-term viability of such computer-assisted improvements.

Wang et al. (2023) [8] formalize the Smart Word Suggestions (SWS) task as an end-to-end writing assistance problem comprising two sub-tasks: improvable target detection—identifying words in a sentence that could benefit from replacement—and substitution suggestion—providing

better alternatives for each detected target. Unlike traditional lexical substitution, which provides a pre-identified target word and seeks meaning-preserving synonyms, SWS requires the system to autonomously detect which words need improvement and then suggest substitutions that enhance writing quality. The benchmark includes 1,000 sentences from English learner essays annotated by 10 native speakers with over 16,000 substitution suggestions, and a distantly supervised training set of 3.7 million Wikipedia sentences. A key finding is that existing baselines achieve at most $F_{0.5}^{e2e} = 0.201$ (BERT) versus human performance of 0.359, motivating the use of structured linguistic resources rather than relying solely on neural generation. The SWS benchmark provides the foundational task definition, evaluation metrics, and test data for this project’s pipeline.

Liu and Henao (2025) [9] address a fundamental limitation of the SWS benchmark: reliance on subjective, incomplete human annotations. They demonstrate that 60.8% of substitution suggestions in the SWS test data were provided by only a single annotator, making ground-truth labels sparse and non-generalizable. To overcome this, they propose using BARTScore—a model-based metric that computes the conditional log-likelihood of a sentence via a pre-trained BART model—as a proxy for sentence quality assessment, eliminating the need for human annotations. Their preference-aware learning framework with margin ranking loss achieves state-of-the-art results, outperforming BERT-SWS, BART-SWS, GPT-4o, and LLaMA across four datasets. This paper provides the theoretical and empirical foundation for incorporating BARTScore as a fluency evaluation component in the proposed pipeline’s ranking stage.

Flor et al. [10] present VXGL (Vocabulary Expected Grade Level), a large-scale mapping of over 126,000 English word forms to U.S. school grade levels ranging from kindergarten (grade 0) through college (grade 16). The resource was built through manual data collection from 74 educational sources yielding grade-level assignments for approximately 22,000 unique word forms, followed by statistical expansion using General Boosting Machines regression with predictors including word frequency, morphological features, POS information, and a novel associative estimation predictor. The VXGL correlates strongly with age-of-acquisition data (Pearson $r = 0.78$) and CEFR L2 levels (Spearman $\rho = 0.647$), suggesting that L1 and L2 vocabulary orderings share a common difficulty progression. This resource is the central differentiating component of the proposed pipeline, enabling grade-aware filtering that neither the SWS baselines nor the BARTScore ranking approach provides, ensuring that suggested vocabulary replacements are both pedagogically appropriate and within reach of the target learner population.

Altogether, the literature review highlights the significance of vocabulary training and the opportunities of AI-based feedback systems. However, no existing approach integrates

real-time lexical intervention, grade-aware filtering, structured synonym generation, multi-model quality scoring, and a strict non-generative constraint within a unified framework. This is where the proposed grade-aware smart word substitution pipeline is motivated.

V. METHODOLOGY

A. System Overview

We propose a grade-aware smart word substitution pipeline for vocabulary coaching of ESL learners. The approach builds on the Smart Word Suggestions (SWS) task formalization [8] and extends it with a WordNet-first candidate generation strategy, VXGL grade-level filtering [10], and multi-model scoring using RoBERTa, BARTScore [9], and SBERT.

The system comprises five neural models and one lexical resource orchestrated in a sequential pipeline:

- WordNet for synonym candidate generation (linguistic resource),
- VXGL for vocabulary grade-level filtering [10],
- RoBERTa-large for contextual reranking via masked language model probabilities [11],
- BARTScore (BART-large, ParaBank2 variant) for sentence-level fluency evaluation [9],
- SBERT (all-MiniLM-L6-v2) for semantic preservation verification [17],
- Mistral-7B-Instruct for optional pedagogical explanation generation.

The system works in six main stages: target detection, candidate generation, multi-criteria filtering, scoring and ranking, iterative multi-word improvement, and explanation generation. Critically, the pipeline adopts a strict non-generative policy motivated by concerns about learner dependency on AI writing systems [20], [21].

B. Problem Formulation

Let a sentence be represented as:

$$S = (w_1, w_2, \dots, w_n). \quad (1)$$

The goal is to identify a subset of words:

$$T \subseteq \{1, 2, \dots, n\} \quad (2)$$

where each word in T can be improved lexically, following the SWS task formulation [8]. For each selected word w_t , we generate synonym candidates from WordNet:

$$C_t = \{c_1, c_2, \dots, c_k\} \quad (3)$$

and choose the best candidate using a composite scoring function:

$$c_t^* = \arg \max_{c \in C_t} \text{SCORE}(c, S, u). \quad (4)$$

C. Stage 1: Target Word Detection

The target detection module identifies words and phrases in the input sentence that are candidates for vocabulary improvement. The function tokenizes the sentence using NLTK word tokenization and POS tagging, then performs two passes:

Pass 1 — Intensifier + Adjective Phrases: The system scans for bigrams where the first word is an intensifier (e.g., “very”, “really”, “extremely”) and the second is a POS-tagged adjective. These phrases (e.g., “very good” → “excellent”) are assigned the highest priority because they represent the most impactful substitution opportunity—replacing a two-word construction with a single more precise word.

Pass 2 — Standalone Content Words: Remaining tokens are evaluated individually. The system skips function words, words shorter than 3 characters, and proper nouns. Adjectives receive the highest priority among standalone words, followed by adverbs and common nouns, reflecting the SWS benchmark finding that adjectives are the most frequently improvable POS category [8].

D. Stage 2: WordNet-First Candidate Generation

A critical design decision in this pipeline is the adoption of a WordNet-first approach rather than the MLM-first approach used in most SWS baselines. The fundamental insight is that masked language models predict contextually probable words, not synonyms. Given a masked sentence, an MLM might predict a contextually coherent but semantically unrelated word. By contrast, WordNet provides guaranteed synonyms, and MLM is used only to rerank among semantically valid candidates—leveraging each component’s strength [12].

For a given target word and POS tag, the function queries WordNet synsets filtered by POS. From each synset, all lemma names are extracted. For adjectives, the function additionally traverses `similar_to_s()` links to capture adjective satellite synonyms that are connected only through these relations rather than shared synsets. Multi-word expressions and short words are excluded.

Candidates are then passed through five sequential filters:

- **Morphological variant check:** Candidates that are sub-strings of the original word or vice versa are removed.
- **VXGL grade-level gate:** The candidate’s grade level must exceed the original word’s by at least 0.3 levels, and must not exceed a maximum grade threshold (e.g., grade 9.0 for middle school ESL contexts) [10].
- **SBERT semantic preservation (≥ 0.80):** The substituted sentence must maintain cosine similarity of at least 0.80 with the original [17].
- **RoBERTa MLM probability gate ($\geq 10^{-5}$):** The candidate’s contextual probability must exceed a minimum threshold to eliminate obscure or archaic synonyms.
- **Sentence identity check:** If the substituted sentence is identical to the original, the candidate is discarded.

E. Stage 3: Multi-Signal Scoring and Ranking

Candidates that survive all filters are scored using a weighted composite of four normalized signals inspired by model-based ranking approaches [9]:

$$\text{SCORE}(c, S, u) = 0.35 \cdot \text{MLMPROB} + 0.25 \cdot \text{BARTSCORE} \\ + 0.20 \cdot \text{SBERTSIM} + 0.10 \cdot \text{GRADEIMPROVE} \quad (5)$$

The scoring integrates:

- **RoBERTa MLM probability (weight 0.35):** Contextual fit via masked language model probabilities, the single most important ranking signal [11].
- **BARTScore (weight 0.25):** Sentence-level fluency evaluation using BART-large to measure how well the substituted sentence can be reconstructed, serving as a proxy for naturalness [9].
- **SBERT similarity (weight 0.20):** Semantic preservation measured as cosine similarity between original and substituted sentence embeddings [17].
- **VXGL grade improvement (weight 0.10):** The grade-level increase from the original word to the candidate, ensuring vocabulary sophistication growth [10].

All four signals are min-max normalized within each target’s candidate pool before weighting, ensuring that the composite score reflects relative quality within the candidate set rather than absolute scale differences between signals.

F. Stage 4: Iterative Improvement and Explanation

The system supports improving multiple words in a single sentence through iterative application. The pipeline runs up to 3 iterations, where each iteration identifies the best remaining substitution and applies it to the already-modified sentence. A tracking set prevents cycles (replacing word A with B, then B back to A) or redundant substitutions.

For each final set of substitutions, the system generates a pedagogical explanation using Mistral-7B-Instruct. The prompt positions the LLM as a writing tutor helping an ESL student, provides the original sentence and a structured list of all changes with their grade levels, and constrains the output to a concise paragraph with supportive tone. Importantly, the explanation module does not rewrite any text—it only explains the vocabulary improvements already selected by the pipeline, maintaining the strict non-generative constraint for the substitution process itself.

VI. EXPERIMENT DESIGN

A. Datasets

The primary evaluation dataset is the **SWS Benchmark** [8], which consists of 1,000 sentences drawn from English learner essays. The test set contains 800 sentences annotated by 10 native speakers, producing over 16,000 substitution suggestions with inter-annotator agreement $PA_{\text{sug}} = 35.4\%$. The benchmark also provides a distantly supervised training set of 3.7 million Wikipedia sentences with 12.7 million

synthetic substitutions generated via PPDB and Merriam-Webster thesaurus. The two suggestion categories defined by the benchmark—*refine-usage* (correcting inappropriate word choices, 63% of targets) and *diversify-expression* (providing vocabulary alternatives)—map directly to the system’s dual goals of grade-level improvement and vocabulary sophistication.

For grade-level filtering, the system uses the **VXGL (Vocabulary Expected Grade Level)** resource [10], which maps over 126,000 English word forms to U.S. school grade levels (K–16). The VXGL dictionary is loaded as a lookup table where each word maps to its expected grade level, with a conservative default of grade 4.0 for unknown words.

Additionally, the **TOEFL11 (ETS Corpus of Non-Native Written English)** [7] may be used as a supplementary evaluation corpus. This large, proficiency-marked corpus of non-native English essays is well-suited for extracting lexical characteristics conditioned by learner proficiency level and for assessing whether the pipeline causes learners’ writing to shift toward patterns observed in higher-scoring texts.

B. Data Preprocessing

For the SWS benchmark evaluation, sentences are processed through NLTK tokenization and POS tagging to identify target words. The benchmark’s gold annotations (target word positions and acceptable substitutions) are used for computing precision, recall, and $F_{0.5}$ metrics. Sentences containing fewer than 5 tokens or consisting entirely of function words are excluded from evaluation.

The VXGL resource is loaded from its CSV format (word, grade columns) into a dictionary structure. Words are lowercased and stripped of whitespace during loading. Grade values are stored as floating-point numbers to enable fine-grained grade difference computations (minimum improvement threshold of 0.3 grade levels).

For neural model inputs, sentences are tokenized using each model’s respective tokenizer: RoBERTa’s byte-pair encoding tokenizer for MLM probability computation (with leading-space handling for subword tokens), BART’s tokenizer for BARTScore computation, and SBERT’s WordPiece tokenizer for semantic similarity. All models operate in evaluation mode with no gradient computation to minimize memory usage and enable efficient batch processing during the scoring stage.

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